INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

September 2025

Subject CS1 – Actuarial Statistics
Core Principles

Paper B

Time allowed: One hour and fifty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

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1 (i) Simulate 200 values from a gamma(6, 0.5) distribution. Use the command `set.seed(1234)` to initialise the random number generator before you start the simulation. [2]

(ii) Explain in words an alternative method for simulating values from a gamma(6, 0.5) distribution based on a suitable exponential distribution. You are not required to implement this method in R. [3]

(iii) Plot a histogram of the values generated in part (i), making sure that the histogram is plotted on the probability density scale. [3]

(iv) Plot the curve of the Probability Density Function (PDF) of the gamma(6, 0.5) distribution on the same graph as the histogram in part (iii). [3]

[**Hint:** Use the `line()` function, or the `curve()` function, with the option `add = TRUE`.]

(v) Comment on the accuracy of the simulation in part (i), using your answers in parts (iii) and (iv). [2]

The R code below, when completed correctly, can be used to simulate values from a Poisson distribution:

```
x = numeric(A)
for (j in 1:A){
  k=1
  u = runif(1,B,1)
  while (ppois(k-1,C) < D){
    k = k+1
  }
  x[j] = E
}
```

(vi) State what **A**, **B**, **C**, **D** and **E** should be replaced with in the R code above, so that the code simulates 1,000 values from a Poisson(5) distribution, which are stored in the vector denoted by x. [5]

[Total 18]

2 An analyst is investigating the delay period between an insurance claim being made and the claim's settlement in a health insurance portfolio. In particular, the analyst is interested in the dependence of the delay period on a number of covariates. A data sample is available in the file 'CS1B_settlement_times.RData', which contains the following variables for each of 200 claims:

- `delay`: the delay period (in weeks) between the claim being made and its settlement
- `age`: the age of the policyholder associated with the claim
- `duration`: the duration for which the policy has been in force (recorded as categorical for analysis purposes, with two levels: low, high)
- `size`: the size of the claim (recorded as categorical for analysis purposes, with two levels: low, high)
- `cause`: the cause of the claim, indicating a certain health condition (categorical with two causes: c1, c2).

The analyst first wants to consider the dependence of the delay period on all four covariates.

(i) Fit a gamma Generalised Linear Model (GLM), using all four covariates with no interaction terms and a logarithmic link. [3]

(ii) Comment on the fit of the model fitted in part (i), using a plot of the deviance residuals of the model against age. [3]

(iii) Comment on the dependence of the delay period on each of the four covariates used in the model fitted in part (i). [3]

(iv) Explain, using the model in part (i), the impact of a 10-year increase in a policyholder's age on the mean delay period. [2]

The analyst then wishes to use a model with fewer covariates.

(v) Perform backward variable selection, based on the GLM fitted in part (i), to select an optimal set of covariates. [9]

[Total 20]

3 An analyst at a medium-sized insurance company has randomly selected 40 claims in an attempt to study if the average claim size for a particular class of insurance policy does not exceed £15,000 p.a. The data is available in the file 'CS1B_ClaimsData.RData', which can be loaded into R using the command `load("CS1B_ClaimsData.RData")`.

The analyst has assumed that claim size is Normally distributed with unknown mean μ and unknown variance σ^2.

(i) (a) Plot a histogram of the data.

(b) Comment on the assumption of Normality. [4]

(ii) Determine a 99% confidence interval for μ based on the 40 claims. [4]

(iii) Test the hypothesis that the expected claim size is greater than £15,000 at the 1% significance level. [6]

(iv) Comment on whether or not there is contradiction between the answers in parts (ii) and (iii). [3]

The analyst is also concerned about the number of claims per policy received each year. The data in file 'CS1B_ClaimNumbers.RData' contains the number of claims for 100 policies for the years 2022 `(NClaims2022)` and 2023 `(NClaims2023)`, and can be loaded into R using the command `load("CS1B_ClaimNumbers.RData")`.

(v) Perform a test to decide if the mean number of claims per policy is the same for both years at the 1% significance level. [6]

The analyst has now begun gathering further information in an attempt to fit a generalised linear model to the claims data. It is now assumed that the data on the number of claims per policy come from a Poisson distribution. Two choices for the link function are being considered, the identity link function and the log link function, respectively.

(vi) (a) State the necessary requirements a link function must have in order to fit a model. [1]

(b) Justify which of the two link functions should be used. [4]

[Total 28]

4 An insurance company is analysing the time until claims are filed for a specific type of insurance policy. The company believes that the time until a claim is filed, X, follows a specific distribution with a fixed parameter k and an unknown parameter λ.

The Probability Density Function (PDF) of this distribution is given by:

$$f(x; k, \lambda) = k\lambda x^{k-1} e^{-\lambda x^k}, \text{for } x \geq 0$$

where:

- $k = 1.5$
- $\lambda > 0$ is the unknown parameter.

For data $x_1, \ldots, x_n$, the log-likelihood function is given by:

$$l(\lambda) = n\ln(k) + n\ln(\lambda) + (k-1)\sum_{i=1}^{n} \ln(x_i) - \sum_{i=1}^{n} \lambda\, x_i^k$$

where n is the number of observations.

The company's historical data of claim times (in months) are given below:

```
data <- c(2.4, 1.1, 3.2, 0.9, 4.7, 1.6, 2.2, 3.8, 2.7,
1.4, 3.3, 0.5, 4.0, 2.0, 1.8)
```

(i) Plot a histogram of the data at density scale. [3]

(ii) Plot the values of the log-likelihood function for the parameter λ based on the historical data. Your plot of the log-likelihood function must be for values of $\lambda = 0, 0.01, \ldots, 1.99, 2$. [6]

The Maximum Likelihood Estimator (MLE) for the parameter λ based on a random sample $X_1, \ldots, X_n$ is given by:

$$\lambda_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} X_i^k}$$

where n is the sample size.

(iii) Estimate the value of λ using the MLE and the company's historical data of claim times given above. [1]

(iv) Comment on the plot in part (ii) using your answer in part (iii). [1]

(v) Plot the PDF $f(x; k, \lambda)$ of X as a function of x for $x = 0, 0.1, 0.2, \ldots, 5$ using the estimated value of λ in part (iii). The graph of the PDF should be superimposed on the histogram produced in part (i). [5]

For the parameter λ of this distribution, the Cramér–Rao Lower Bound (CRLB) is given by:

$$\text{CRLB} = \frac{\lambda^2}{nk^2}$$

(vi) Calculate the value of the CRLB using your estimate for λ from part (iii). [3]

An approximate confidence interval for λ can be obtained by using the approximate distribution of the MLE, $\widehat{\lambda}$, which is a normal distribution with mean λ and variance given by the CRLB.

(vii) Calculate an approximate 95% confidence interval for λ using the normal distribution above and your answers from parts (iii) and (vi). [2]

The company wants to incorporate prior knowledge about the parameter λ and assume that the prior is a gamma distribution with parameters $a = 12.6$, $b = 30$.

The posterior distribution of λ follows a gamma distribution with parameters $a + n$ and $b + \sum_{i=1}^{n} x_i^k$.

(viii) Calculate the values of the parameters of the posterior distribution of λ. [3]

(ix) Calculate the Bayesian estimate of parameter λ under the quadratic loss function. [2]

(x) Simulate 10,000 values from the posterior distribution of λ and store the simulated values in a vector in R for later use.

You should use the command `set.seed(123)` to initialise the random number generator before you start the simulation. [3]

(xi) Determine a 95% credible interval for λ using your answer in part (x). [3]

[**Hint:** You may find the R command `quantile()` useful.]

(xii) Comment on the intervals in parts (vii) and (xi). [2]

[Total 34]

END OF PAPER